

**TECHNOLOGICAL ACCEPTABILITY OF A PREDICTIVE CHEMICAL
TOXICITY SOFTWARE FOR SCIENCE-RELATED PROGRAMS
IN A CATHOLIC SCHOOL IN DAVAO CITY**

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May this research serve as an inspiration that inquiries like these are not made for compliance but for advancing our position as Filipino citizens in the rapidly changing landscape of globalization. Indeed, ***innovation starts when we learn to observe, act, and pursue.***

- The Researchers

ABSTRACT

Toxicity awareness in undergraduate laboratory settings remains low in the status quo, as students often avoid reading physical manuals that detail the risks of the substances they use. To combat this, online software predicting chemical toxicity has been developed, such as ProTox 3.0, detailing classifications, consequences, and reports when in contact with humans. However, research on its effectiveness and user satisfaction is scarce, especially from a geographical and empirical perspective. Hence, this descriptive-comparative study aimed to determine the level of technological acceptability when using ProTox 3.0 among college students in an institution in Davao City, under programs that utilize scientific laboratories and handle chemical reagents. Through an online adapted survey questionnaire that gave respondents a brief overview of the software, distributed among 121 1st to 3rd year students studying in science-specialized programs, the analyzed results showed a high level of technological acceptability among the respondents, with a relatively significant difference based on their sex. These findings highlighted the need for enforcing innovative methods to deliver crucial safety information among students, as it proves effective regardless of their age, chosen program, or year level. Similarly, incorporating digitalization into safety awareness may include gender-neutral delivery to cater to the acceptability differences between users' sexes.

Keywords: *ProTox 3.0, predictive chemical toxicity software, quantitative, Philippines, descriptive-comparative, technological acceptability*

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INTRODUCTION

Background of the Study

Many students in the laboratory struggle to distinguish the harm and toxicity of the chemicals they use during experiments; as a result, they may mix or use chemicals without fully knowing their risks, and this lack of information and awareness can lead to accidents before they even begin. In today's era, however, technology plays a vital role in making tasks faster and more efficient, increasing laboratory awareness among individuals, especially in education.

In other nations such as the United States of America, a U.S government agency known as the EPA, studies issues of predictive chemical toxicity software reliability compared to animal test or human data, and if the model is able to handle all kinds of chemicals, or only the ones similar to what it was trained on. According to the study of the U.S. Environmental Protection Agency (2020), agencies have expressed concern over the generalizability of predictive models outside their training chemical space and have been cautious in using in silico models. Despite this, the EPA's New Approach Methods is moving towards the future of Chemical Assessments and supportive in principle of in silico approaches. However, they have been slow to adopt them as standalone evidence, particularly for high-stakes assessments.

The Department of Environment and Natural Resources (DENR) and Environmental Management Bureau (EMB) located in the National Capital Region of Quezon City, regulates chemicals under the Philippine Chemicals Control Order and Toxic Substances and Hazardous and Nuclear Wastes Control Act RA 6969. Although the Philippines does not yet have its own predictive software, such as EPA's ToxCast, it often adapts to international frameworks similar to EPA and OECD. EMB accepts undergraduate interns and assigns them to laboratories, monitoring divisions, or permitting offices, where they learn about environmental sampling, chemical safety, and regulatory processes. They often get involved in chemical waste inventory projects, environmental

monitoring, and education campaigns. The students also assist in updating databases by encoding chemical data and conducting literature reviews of known chemical hazards for DENR reports. After graduating, some students often get placed in DENR/EMB projects, creating an indirect pipeline between undergraduate work and agency research. Unlike the U.S. Environmental Protection Agency, which actively involves undergraduates with predictive toxicology tools, Philippine institutions tend to rely on existing—often foreign—frameworks. There is a lack of locally developed software and dedicated research into how these tools affect learning in the Philippine academic context (Banerjee et al., 2024; Bhatnagar et al., 2024).

One of the locale studies that is interconnected to region XI is the Poison Control Network (PCN) is developing an information system, Toxicology Information System (TIS), which aims to interoperate the operations of 13 poison control centers in the Philippines in order to improve the quality, monitoring, outcome, safety, and/or effectiveness of poison control and treatment of poisoning cases. The system is accessible to authorized persons daily with appropriate data sharing functionalities (i.e., care team, public dashboard, and research portal) while maintaining patient data privacy and security. The system will also serve as the centralized source of toxicology patient data of the 13 poison control centers for research purposes and a pesticide surveillance system so that the health (in terms of pesticide/chemical poisoning) of farmers and factory workers is monitored (PCN-TIS, 2025).

ProTox 3.0 is a virtual lab for the prediction of toxicities of small molecules. The prediction of compound toxicities is an important part of the drug design development process (Priyanka et al., 2018; Banerjee et al., 2024). Computational toxicity estimations are not only faster than the determination of toxic doses in animals and humans, but they can also address the issue by helping to reduce the amount of animal experiments and how they affect humans. According to Banerjee et al. (2024). The successful implementation of this kind of technology depends on factors outlined in technology acceptance models. These include how easy students find the software to use, how useful they perceive it to be, any

technical drawbacks (such as the range of chemicals covered or the limitations of the models), and, of course, student opinions (Davis et al., 1989; Chandrasekaran et al., 2025).

From the studies mentioned above, there exists a knowledge gap in investigating the use of predictive chemical toxicity software within an academic college setting, specifically on its effects with science-oriented undergraduate students. This study will address this knowledge gap by looking into the students' perspectives and seeing firsthand the potential this software has within the classroom. In addition to this, there is a population gap. As stated, there are few to no research papers investigating the use of this software in college, and so college students are an untapped group of respondents.

This study investigates how predictive chemical toxicity software can help students in science-related programs by making their tasks easier and improving their understanding of different chemicals. While software like ProTox 3.0 holds promise for speeding up research, improving safety, and conserving lab resources, especially in fields like chemistry, its use in Philippine undergraduate science education hasn't been thoroughly investigated. Specifically, it aims to determine how such software can support students in completing academic tasks more efficiently and deepen their understanding of chemical concepts. The results of this study are expected to help students adjust to using modern scientific tools and guide schools in improving the use of technology in science programs.

Therefore, to share the findings of this research, the researchers shall ensure the following dissemination measures for the respective audiences. After the defense and any necessary revisions, the results will be presented to students, teachers, and faculty members at the catholic university through posters and class presentations. Once approved, the study will be shared during a school-based research forum so that other students in science-related programs benefit from the study. If possible, the research may also be uploaded to the university digital platforms or library for future reference and may later be considered for publication in

online research journals or websites such as Academia.edu and Google Scholar. All these steps will be guided by feedback from experts and the group's research adviser to ensure that the study is clear, reliable, and relevant to the academic community.

Statement of the Problem

This study aims to determine the levels of technological acceptability when using predictive chemical toxicity software among first- to third-year students under science-related undergraduate programs. Furthermore, it also measures if there exists a significant difference between the level of technological acceptability among first- to third-year students.

Specifically, it seeks to answer the following questions:

1. What is the demographic profile of the respondents in terms of:
 - 1.1 age bracket;
 - 1.2 sex;
 - 1.3 year level; and
 - 1.4 program?
2. What is the level of technological acceptability of the respondents in terms of:
 - 2.1 autonomy;
 - 2.2 competence;
 - 2.3 relatedness;
 - 2.4 perceived ease of use;
 - 2.5 perceived usefulness; and
 - 2.6 behavioral intention to use?
3. Is there a significant difference in the technological acceptability among respondents when grouped according to age bracket, sex, year level, and program?

Hypothesis

Null Hypothesis (H_0): There is no significant difference in students' level of technological acceptability when grouped according to age bracket, sex, year level, and program.

Alternative Hypothesis (H_a): There is a significant difference in students' level of technological acceptability when grouped according to age bracket, sex, year level, and program.

Significance of the Study

This study aims to determine the levels of technological acceptance on the utility of predictive chemical toxicity software among first- to third-year students. The possible results of this study could provide benefits to the following beneficiaries:

Students. This study will help students understand the importance and applicability of predictive chemical toxicity software to enhance their knowledge of computational toxicology, essential for modern scientific research.

Faculty. This study provides educators insights into students' acceptance and adaptability towards integrating predictive chemical toxicity software into the curriculum. Educators can also utilize these findings to improve teaching methods, implement technology-based activities, and prepare students for advanced scientific roles.

The Catholic School and Institutional Stakeholders. This study provides the institution with valuable insights into adopting advanced technological tools for enhancing science education and research capabilities in line with modern trends.

Researchers and Developers of Predictive Chemical Software. Insights from the study could help software developers understand the

user acceptance challenges and requirements from an academic perspective, facilitating improvements and wider adoption.

SDG 4. The skills of a chemistry graduate are in high demand not only for employment in chemistry but in many sectors in which analytical thinking is required, which is also significant to the 4th Sustainable Development Goal issued by the United Nations (2015), which aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Scope and Limitation

The focus is primarily on the level of technological acceptability regarding the use of predictive chemical toxicity software as part of the students' academic learning and exposure to scientific tools. The scope is limited to the educational impact and acceptance within the institution and does not evaluate the software's technical accuracy or toxicological validity. Limitations include the specificity of the student population, inherent challenges of the software, such as prediction accuracy and model complexity, and varying influences from students' prior technological exposure, curriculum, or teacher support.

Furthermore, due to time constraints, some significant parts of the research process were bypassed to cater to its punctual completion, such as survey validation, reliability testing, ethical clearance, and guidance from a licensed statistician. Regardless, the researchers are well-informed of the process, and ensured that ethical considerations and due conduct of data analysis have been followed thoroughly to minimize potential quantitative errors in data.

Review of Related Literature

Understanding trends in predictive chemical toxicity software often involves findings from other sources that detail user insights, acceptability, and impacts on laboratory safety. In this chapter, the researchers detailed how varying related studies may align with the current perspective on the

research topic, emphasizing paradigms in similar software acceptability via other conditions, whether geographical or circumstantial.

Toxicity Awareness in Laboratory Settings

According to a study conducted by Zaip et al. (2021), laboratory workers in Universiti Sains Malaysia Health Campus scored a moderate overall level of chemical safety awareness, where element chemical management had the highest mean score of 3.74. This moderate level showed awareness of the storage of chemicals by toxicity level, and by extension, chemical toxicity.

A study by Papadopoli et al. (2020) found that researchers, although aware of most investigated chemical hazards and how they may affect health, are not very confident in how to protect themselves, primarily due to insufficient knowledge about personal protective equipment (PPE). Younger researchers and those with longer laboratory experience demonstrated significantly higher knowledge. Additionally, researchers who work with a larger number of chemical compounds tend to have better knowledge, indicating that direct experience with chemicals increases awareness of exposure risks.

Predictive Software as a Precursor to Actual Laboratory Performance

A study by Bhat and Chatterjee (2021) explored the use of toxicity prediction servers in finding potential inhibitors for SARS-CoV-2. Three servers were named: Protox-II, DL-DILI Prediction Server, and VEGA. 19 chemical compounds, which were found to potentially inhibit SARS-CoV-2, were studied using these servers, namely for their acute oral toxicity, carcinogenicity, hepatotoxicity, and other potential adverse effects. In this study, no true testing within a laboratory was done. This showed predictive software as a way of reducing costs and accelerating experimentation through weeding out the less favorable compounds in developing drugs.

Laboratory Users' Insights on Using Software for Laboratory References

Recent studies reveal growing interest in using digital and virtual tools in chemistry education to enhance students' understanding and engagement. The "Crocodile Chemistry" virtual lab program, for example, has been effective in improving motivation and learning by allowing safe, repeatable experiments before physical labs (Alqallaf et al., 2025). Meta-analyses support virtual labs' positive impact on engagement and comprehension (Castro, 2025), while other studies confirm their role as effective supplements or alternatives to traditional labs (Bazie et al., 2024).

In the Philippines, students using computer-based simulations like PhET showed increased knowledge, interest, and independence in chemistry learning (Asedillas & Quimbo, 2019). Exposure to such simulations correlates strongly with better conceptual understanding and retention, highlighting their value in addressing complex topics (Fabrigas & Paglinawan, 2025; Batamuliza et al., 2024). Students generally find these tools motivating and user-friendly, facilitating independent learning.

Research on students' perceptions of laboratory software and technology reveals mixed findings across different contexts. Thomas et al. (2004) examined students' early experiences with microcomputer-based laboratories (MBL), providing insights into initial user reactions to laboratory technology integration. Morris and Dillon (1997) demonstrated that technology acceptance models focusing on system usefulness, beyond traditional usability measures, significantly influence user acceptance of software systems. Teng et al. (2016) investigated student perceptions of remote laboratories through a case study of NetLab, finding that both academics and students increasingly accept properly structured remote laboratories as valid complements to or replacements for traditional physical laboratories.

Feasibility of Predictive Chemical Toxicity Software in Practicing Laboratory Safety

According to the study of Stucki et al. (2022), New Approach Models (NAMs) can be utilized individually or in combination to improve chemical safety assessment through more protective and relevant models

without the use of testing animals. Adopting NAMs has the potential to result in significant scientific advancements as well as, in certain cases, financial gains. The quantitative structure-activity relationship is one of several computational toxicology methods that predict toxicity based on a chemical substance's structure and are attracting increasing interest. With the use of big data-driven artificial intelligence (AI) technologies, computational toxicology in particular is set for future advancement.

In a recent study on toxicity prediction models based on artificial intelligence (AI), researchers' first option for creating QSAR-based prediction models is the ToxCast data (Jeong & Choi, 2022). The primary reasons why ToxCast data is preferred for creating AI models are the large volume of data and the high level of homogeneity brought about by standardized data processing (Feshuk et al., 2023). A significant amount of data for model creation is provided by ToxCast data, which includes publicly accessible in vitro screening findings for thousands of compounds across many assay platforms.

Definition of Terms

In this section, the researchers aim to provide a common conceptual basis by providing readers with carefully prepared terminology that will guide them throughout the study.

Technological Acceptability. Technology acceptability refers to the behavioral outcome predicted by perceived usefulness, ease of use, and intention to use, indicating how likely a user is to adopt and use a technology when they find it useful and easy to operate.

Predictive Toxicity. Predictive Toxicity refers to the prediction of potential toxic effects of a chemical or drug on living organisms, including humans, as well as the safety and potential risks associated with exposure to chemicals, drugs, environmental pollutants, and other substances.

Predictive Chemical Toxicity Software. Predictive chemical toxicity software refers to computational tools that utilize mathematical

models, including Quantitative Structure-Activity Relationship (QSAR) models and machine learning algorithms, to predict the toxic effects of chemical compounds without physical testing.

Theoretical Framework

Self-Determination Theory (SDT) serves as an extensive framework for examining human motivation and personality. It provides a meta-theory for organizing motivational research, a formal theory that outlines intrinsic and diverse extrinsic motivations, and an explanation of the roles that intrinsic and extrinsic motivation play in cognitive and social growth, as well as in individual variations (Deci & Ryan, 1985). These needs consist of **autonomy**, **relatedness**, and **competence**, which nurture intrinsic motivation and encourage optimal learning and performance. In this study, Self Determination Theory helps to explain the motivational factors that impact students' technological acceptability of predictive chemical toxicity software.

In recent Technology Acceptance Model and Self Determination Theory integration (Racero et al., 2020), behavioural intention has been shown to be a reliable determinant of technology acceptance among secondary school students. When learners perceive a sense of autonomy in their learning, believe in their competence to master the software, and experience relatedness through support and connection with peers and teachers, they are more likely to accept and apply the technology in their academic environment.

Autonomy

Individuals must have the perception that they can influence their own actions and objectives. This ability to implement meaningful change significantly contributes to promoting a sense of self-determination in people (Cherry, 2024). In this study, autonomy is shown in students' freedom to explore and widen the use of the predictive chemical toxicity software that matches their learning style. When students feel that using the software is not only applied but also supports their learning, they are

more likely to view it as both beneficial and easy to use. A study that combined Self-Determination Theory and the Technology Acceptance Model found that autonomy positively influenced students' perceived usefulness and perceived ease of use of educational software, which in turn increased their behavioral intention to use it (Racero et al., 2020).

Relatedness

Relatedness refers to the capacity to experience attachment to others as well as a sense of belonging within a group. The feelings of relatedness become stronger when individuals feel valued and supported by those around them and are included in a welcoming environment, whereas they can be diminished by rivalry, exclusive groups, and negative feedback from others (Lopez-Garrido, 2023). In this study, relatedness is important in students' ability to use the predictive chemical toxicity software as a tool for discussion, communication, and collaboration in science-related courses. When students feel valued and supported with the software, their motivation to accept and keep using it increases. This relates to Chiu (2024), who discovered that relatedness had a positive effect on students' perceptions of the usefulness and ease of use of educational software, which subsequently enhanced their intention to engage with it.

Competence

Competence signifies the requirement to feel efficient in achieving desired results and in mastering tasks and challenges. Within educational environments, competence has been demonstrated to be a key factor in students' motivation, as individuals who view themselves as capable tend to be more engaged and eager to learn (Wang et al., 2019). If the students feel confident that they can analyze and apply the outcomes, their understanding of their usefulness and ease of use will increase. In addition, previous studies on education that integrated Self-Determination Theory (SDT) and Technology Acceptance Model (TAM) in extracurricular learning environments found that individuals' perception of their competence was the key factor affecting both the perceived usefulness

and perceived ease of use, which subsequently influenced learners' intentions to keep using the technology (Fathali & Okada, 2018).

Conceptual Framework

Within the context of the study, the researchers aim to determine how students' demographic background and personal experiences shape their level of technological acceptability when utilizing predictive chemical toxicity software, ProTox 3.0. By identifying patterns across different age groups, sexes, year levels, and programs, the researchers can understand whether these demographic factors influence the way students perceive, evaluate, and adopt digital tools in their academic learning. Nevertheless, these demographic variables are not measured as direct causes; they provide important grouping, categories that may uncover differences in students' willingness to accept and use new technology.

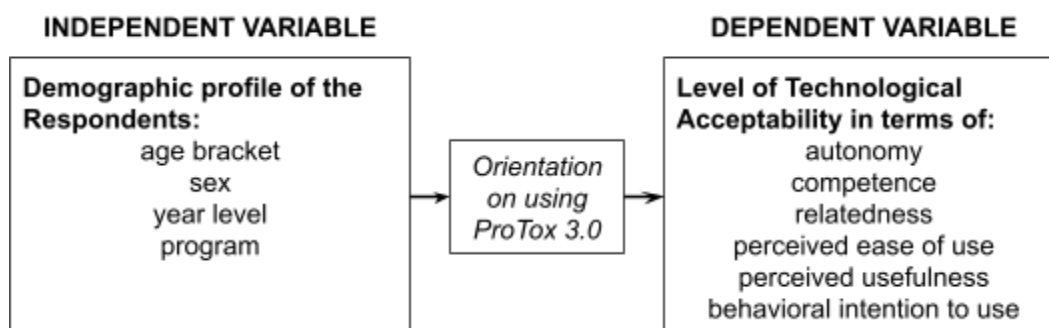


Figure 1. Conceptual Framework of the Study

Understanding the study's independent variable, the demographic profile of the respondents refers to the basic characteristics that distinguish one student group from another. This profile is measured through four indicators. The first indicator, **age bracket**, refers to the age range of the respondents, which may affect their exposure and easeness with the new technologies (Merchant et al., 2015). The second indicator, **sex**, refers to whether the respondents identify as male or female, recognizing that gender differences may sometimes reflect varying

perspectives in technology adoption (Iddrisu et al., 2025). The third indicator, **year level**, points to whether the student is a first, second, or third year undergraduate, acknowledging that students' academic maturity and course may influence their openness to technology (Staddon, 2020). Lastly, the **program**, which refers to the respondents' enrolled science-related undergraduate program (e.g., Biology, Chemistry, Nursing, Pharmacy, etc.), as the nature of their chosen academic discipline may shape how relevant and useful they perceive computational tools to be in their academic work (Qu et al., 2024).

Meanwhile, the study's dependent variable, the level of technological acceptability, pertains to how students recognize, evaluate, and intend to use ProTox 3.0 in their learning experience. This variable involves six indicators. The first indicator, **autonomy**, refers to the degree of freedom and self-direction the respondents feel when using the software, signifying if they view its use as self-motivated rather than imposed. The second indicator, **competence**, refers to the respondents' perceived ability to effectively operate the software, showcasing their confidence in applying it to complete their tasks. The third indicator, **relatedness**, refers to the respondents' sense of connection and social belonging through the software, such as feeling supported by peers or instructors when using it. The fourth indicator, **perceived ease of use (PEU)**, is how effortless and simple the respondents find the software to operate, underlining its user-friendliness. The fifth indicator, **perceived usefulness (PU)**, refers to the respondents' belief that the software enhances their academic performance by making tasks more efficient. Lastly, **behavioural intent to use** refers to the respondents' expressed willingness and plans to continue using the application, ProTox 3.0, in future academic endeavors, serving as a strong indicator of actual usage.

By examining these indicators, the researchers seek to capture both the personal and technological dimensions of students' acceptance of ProTox 3.0. Demographic factors serve as grouping variables that may explain variations, while the six technological acceptability indicators provide a comprehensive view of how students perceive, experience, and intend to engage with this emerging digital tool.

METHODOLOGY

Research Design

This study used a quantitative method, particularly the descriptive-comparative research design. In academic studies, the quantitative approach is an empirical system of deriving observations through numerical and statistical results to explain a phenomenon (Taherdoost, 2022).

The descriptive component of the design involves systematically documenting the characteristics of variables as they naturally occur (Ansari et al., 2022). Comparative research, then, requires the researchers to compare two or more groups that are exposed to a cause or the variable that they possess, thus determining significant differences between the demographic characteristics of the respondents during the conduct and analysis of the study (Creswell, 2018).

Integrating them altogether, the descriptive-comparative research design, as defined by Devi (2022), is made to compare the outcomes that multiple groups may experience within the environment and scope of the study, allowing the researchers to characterize their differences through empirical analysis. Through its associated statistical tools, the researchers find this structure of the research design to be the most appropriate to use, since it lays a foundation for understanding ProTox's mechanics and usability among different groups of undergraduate students, identifying key similarities and differences that the responses will generate.

Research Locale

This study was situated in Mindanao, specifically in Davao City. Although the city serves as a crucial hub for resources, the larger island has been affected by socio-economic issues like poverty, national conflicts, and corruption (United States Agency for International Development, 2020). Nevertheless, there is a significant need for academic research in this area, offering an important context for this study.

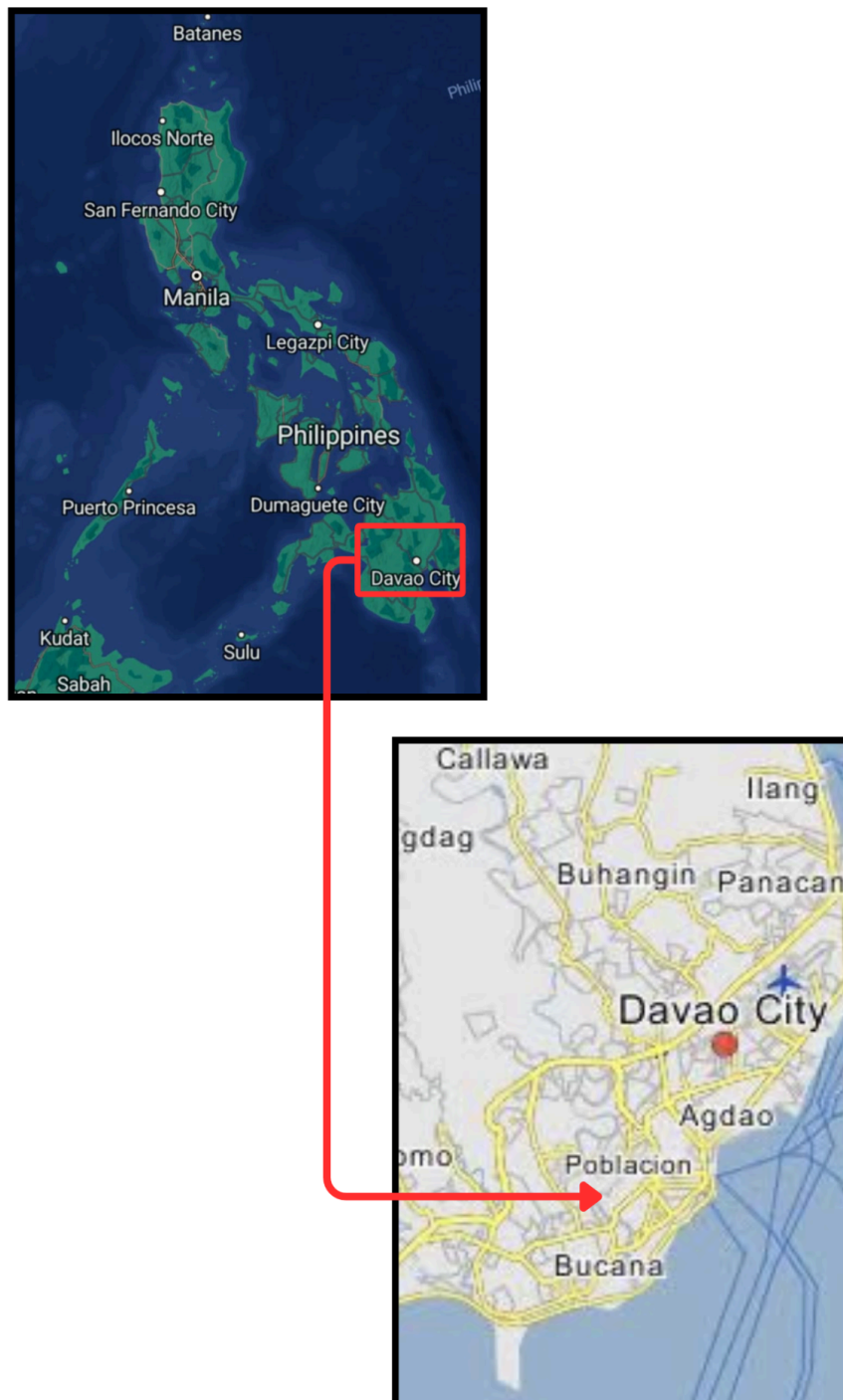


Figure 2. Maps of the Philippines and Davao City

Specifically, the researchers utilized an established university as the study's main locale. As one of the pioneering Catholic schools in Davao City, this institution is recognized for its strong foundation in teaching science-related programs and its high standards of quality education, reflected in commendable board examinations passing rates. Further, this school offers curricula for higher education, offering various pre-medical college programs focusing on medical, natural, physical, and environmental sciences. Aligned with this, the school is known for producing alumni anchored in the integration of the United Nations' Sustainable Development Goals (SDGs), Laudato Si, and the Framework on Justice and Peace, Ecological Integrity, Engaged Citizenship, Poverty Alleviation/Reduction, Gender Equality, and Youth Empowerment (JEEPGY). Lastly, this institution provides scientific and medical facilities, such as laboratories, to ensure that students are immersed in the scientific process to elaborate and explain concepts relevant to their field.

Research Respondents

The researchers included 121 respondents who are bona fide students of the locale. Mainly, the investigation focused on the students enrolled in the following programs: BS Biology, BS Chemistry, BS Medical Laboratory Sciences, BS Nutrition and Dietetics, BS Nursing, BS Pharmacy, and BS Psychology. Additionally, the respondents were also in 1st to 3rd Year standing in terms of their academic performance at the time the study was conducted. Students who did not meet these program or year-level criteria were excluded from the study.

The researchers utilized convenience sampling, a non-probability method that recruits willing and eligible respondents who are readily accessible. This approach was the most suitable given the study's time constraints, as it permitted efficient data collection by avoiding the more complex and time-consuming processes required by other sampling techniques.

Research Instruments

To ensure that the respondents begin the study with a leveled understanding of the research topic, the researchers provided an instructional video that outlines the use of ProTox 3.0 and how the user can navigate its interface. This video included the purpose of the software as well as chemical selection and basic analysis. Afterward, the study utilized an adapted survey questionnaire with two parts. The first part pertained to the respondents' demographic profile, which involves their age bracket, sex, year level, and respective college program.

The second part measured the levels of technological acceptability on the use of predictive chemical toxicity software among the respondents, from the study of Racero et al. (2020), titled "Predicting Students' Behavioral Intention to Use Open Source Software: A Combined View of the Technology Acceptance Model and Self-Determination Theory". This tool consisted of six indicators, namely autonomy (four questions), competence (five questions), relatedness (four questions), perceived ease of use (three questions), perceived usefulness (three questions), and behavioral intention to use (three questions), which can be answered using the four-point Likert scale, where choosing **4** refers to "**Strongly Agree**", **3** refers to "**Agree**", **2** refers to "**Disagree**", and **1** refers to "**Strongly Disagree**".

Range Interval Descriptive Rating		Interpretation
3.26 - 4.00	Very High	The students' level of technological acceptability is always evident.
2.51 - 3.25	High	The students' level of technological acceptability is often evident.
1.76 - 2.50	Low	The students' level of technological acceptability is sometimes evident.
1.00 - 1.75	Very Low	The students' level of technological acceptability is never evident.

Overall, the survey questionnaire was a 22-item instrument evaluating the respondents' technological acceptability when using ProTox 3.0. To ensure the research instrument was suitable for the study's use,

the survey questionnaire was contextualized in line with the research topic, adjusting the grammar, syntax, and structure of each item to ensure that no item would repeat and only ask one item from the respondents.

Data Gathering Procedure

To initiate the data-gathering process, the researchers obtained formal permission from their institution's program deans through a letter, which allowed them to conduct the study and gather data from students within the school premises. All ethical considerations, such as the privacy and confidentiality of the respondents, were ensured. Once approval was obtained, an Informed Consent Form (ICF) was provided to respondents before the survey, describing the study's purpose and procedures during the designated data-gathering period. To maximize the exposure of the survey, the researchers utilized online social media platforms to invite potential respondents to answer the survey. A brief orientation video was provided, and respondents proceeded with the online survey after providing their consent.

Clear instructions guided respondents while answering the survey, and the responses were collected once they submitted their responses. During data collection, the data were monitored and sorted according to validity to ensure completeness and accuracy. As such, any incomplete responses were excluded, and supplementary respondents were recruited to maintain the target sample size. Finally, all valid data collected were compiled and analyzed using appropriate descriptive and statistical methods, ensuring logical consistency in the results for each item.

Statistical Tools

The researchers utilized the following statistical tools to gather, analyze, and compare the data collected using this approach and determine whether significant differences exist among respondents based on their demographic characteristics.

Frequency Counts and Percentage. These are fundamental descriptive statistical tools used to summarize both categorical and quantitative data, providing a clear overview of data distribution. Frequency counts indicate how often a particular response, value, or category appears in a dataset, while percentages express these counts as proportions of the total, facilitating easier comparison across groups. These provide an organized summary of the respondents' characteristics, forming the basis for grouping further statistical comparisons.

Mean and Standard Deviation (SD). The mean is calculated as the total sum of data points in a sample divided by the total number of elements. This tool allows the researcher to determine the respondent's level of technological acceptability by identifying the average score for each item on the survey. Standard deviation, on the contrary, measures the quantity of dispersion of a data set related to its mean. A low standard deviation will indicate that responses were clustered close to the average, while a high standard deviation reflects greater diversity in the respondents' perspective.

T-test. This is an inferential statistical procedure employed to determine whether a significant difference exists between the means of two groups and their relationship to each other. It facilitates hypothesis testing in statistics by utilizing the t-statistic, t-distribution values, and degrees of freedom for assessing statistical significance. When analyzing the average of a sample of measurements, this is the most frequently used technique for data evaluation.

Analysis of Variance (ANOVA). This is an inferential statistical technique to determine whether there are significant differences between the means of three or more independent groups. It is utilized to test hypotheses about group differences without inflating the risk of Type 1 errors, which can occur when multiple t-tests are used instead. By comparing the variance between and within groups, the ANOVA determines if at least one group's average response significantly differed from the others. This will help researchers understand whether perceptions toward Protox 3.0 varies across disciplines and age groups.

Ethical Considerations

Ethical integrity was utilized throughout the design and implementation of this study. The participants were informed beforehand of the study's objectives, potential risks, and their right to withdraw without penalty. In line with this, the researchers adhered to the following principles:

Social Value. Given the growing role of technology in education, effective digital tools are essential for improving scientific learning. Catholic institutions can further strengthen science education by integrating innovative software while upholding the values of ethical responsibility and stewardship as emphasized in *Laudato Si'*. Moreover, focusing on holistic development ensures that technology supports both academic excellence and moral formation. Furthermore, predictive software can be a valuable aid in teaching chemical safety and environmental responsibility. Therefore, this study seeks to enhance science education and develop strategies for effectively integrating technology into the curriculum.

Informed Consent. This involves the permission granted by the program deans and the research adviser to conduct the study. Informed consent was the basis for the participants' ethics in the research. Moreover, respondents received detailed knowledge of the study's aims, coverage, and techniques regarding the use of predictive chemical software for assessing toxicity via orientation. Afterwards, the respondents decided to participate only after all possible questions for clarification have been answered and the scope of the study was explained in detail.

Vulnerability of Research Respondents. The participants in this research were undergraduate students; hence, balancing privacy concerns and vulnerabilities that are personal and technological was done with care. Data anonymization and limiting access to the raw data to only the authorized researchers were some means of addressing data privacy. This particular research only collected students' attitudes and behavioral

acceptance levels, and did not attempt to gather sensitive personal identifiers beyond the disclosed academic information.

Risks, Benefits, and Safety. The participants were exposed to minimal risk from the survey and evaluation instruments associated with software use and its effectiveness, and thus, should not inflict much physical or psychological harm. Information was collected and processed from all participants during the procedures, which were carried out regarding the respondents' privacy and the principles of safe and responsible research. Students who participated in the research will also be provided with a summary of the findings and will, thus, be informed how their participation advances academic and technological improvements.

Privacy and Confidentiality of Information. Protecting the privacy and confidentiality of the respondents' personal information is important in this study. All the collected data, including their personal information and responses, were securely hidden to ensure the confidentiality of the information. The collected data was restricted to members of the research team only, and the results will be presented in a manner that cannot reveal their individual identities. Therefore, the researchers guarantee that no personal data will be leaked or shared with a third party in generating findings for this study.

Justice. Justice ensured that all participants, from first to third year students, had equal access to the predictive chemical toxicity software and the opportunity to explore its applications, regardless of academic level or background. In this study, no individual was treated differently or discriminated against; rather, every student was encouraged to engage freely with the software and maximize its educational benefits. The researchers strongly promoted an environment where learners could enhance their understanding of chemical toxicity predictions without fear of bias, limitations, or exploitation. Ultimately, justice in this context upholds fairness, inclusivity, and a collective commitment to advancing technological competence and scientific learning.

Transparency. Transparency was practiced in the data collection process. The researchers conducted a thorough orientation about the details of this study, such as the purpose, the goals, the risks, and the relevant stakeholders that the participants will take part in.

Qualification of the Researcher. The researchers are apt and experienced with ethical research procedures to execute the study responsibly. Their expertise in survey design, data gathering, and data analysis ensured proper ethical standards and accuracy in the findings. Moreover, the researchers were accompanied by advisors well-versed in the field to corroborate and facilitate the researchers' ethical practices.

Adequacy of Facilities. The researchers ensured that all essential facilities were available, including safe and suitable locations for conducting surveys and interviews, secure data storage solutions, and resources to support participants, such as access to computers, internet connectivity, and the predictive chemical toxicity software needed for the study.

Community Involvement. This research oriented the values of a Catholic school community and recognized key inputs from fields in medical research, healthcare, toxicology, chemistry, and therapeutics. This involvement ensured the research objectives were aligned with community benefit. In addition, this study aligns core values within the United Nations Sustainable Development Goals (UN SDGs) under SDG 3: Good Health and Well-Being, SDG 4: Quality Education, and SDG 12: Responsible Consumption and Production along with the Laudato Si Goals (LSGs) under LSG 5: Ecological Education and Global Compacts on Education (GCE) under GCE 1: To make the human being the center. Subsequently, the study's findings can be applied to support efficient learning and insights among the school and the community.

RESULTS AND DISCUSSION

This chapter presents the necessary data and quantitative analysis needed to answer the study's research questions. Particularly, it gathers information on the respondents' demographic profile, their level of the research variable, and the empirical data that proves whether or not a significant difference is present when grouped according to their characteristics. Hence, the researchers provide the following tables.

Demographic profile of the respondents

Analyzing the respondents' demographic profile in Table 1, the researchers used frequency counts and percentages to represent the number of people who belong to their corresponding profile in their age bracket, sex, year level, and college program. In terms of their age bracket, the majority of the respondents (79.3%) fell within the bracket of 17-18 years old, followed by 19-20 years old (17.4%) and 23 years old and above (2.5%), with respondents aged 21-22 years old having the lowest count (0.8%). Furthermore, a higher percentage of females (80.2%) than males (19.8%) was recorded during the survey.

Most of the respondents were also 1st year college students (89.2%), compared to 2nd year (8.3%) and 3rd year students (2.5%). Lastly, when grouped by programs, the largest proportion of students who took the survey belonged to the Bachelor of Science in Medical Laboratory Science (BSMLS) with 60.3%, followed by Bachelor of Science in Pharmacy (BSPH) (21.5%), Bachelor of Science in Nursing (BSN) (7.4%), Bachelor of Science in Chemistry (BSCh) (4.2%), Bachelor of Science in Nutrition and Dietetics (BSND) (3.3%), Bachelor of Science in Psychology (2.5%) and Bachelor of Science in Biology (0.8%).

Table 1.*Profile of the respondents*

Age Bracket	Frequency	Percentage
17-18 years old	96	79.3%
19-20 years old	21	17.4%
21-22 years old	1	0.8%
23 years old or above	3	2.5%
Total:	121	100%

Sex	Frequency	Percentage
Male	24	19.8%
Female	97	80.2%
Total:	121	100%

Year Level	Frequency	Percentage
1st Year	108	89.2%
2nd Year	10	8.3%
3rd Year	3	2.5%
Total:	121	100%

College Program	Frequency	Percentage
Bachelor of Science in Biology	1	0.8%
Bachelor of Science in Chemistry	5	4.2%
Bachelor of Science in Medical Laboratory Science	73	60.3%
Bachelor of Science in Nursing	9	7.4%
Bachelor of Science in Nutrition and Dietetics	4	3.3%
Bachelor of Science in Pharmacy	26	21.5%
Bachelor of Science in Psychology	3	2.5%
Total:	121	100%

Level of technological acceptability of the respondents

Based on the results, the standard deviation across all categories ranged from 0.60 to 0.83, which was less than 1.0, indicating consistency in the respondents' answers. The overall mean for technological acceptability level was 3.11, classified as "High." This indicates that ProTox 3.0 is well-accepted among undergraduate students, suggesting that this technology could be effectively integrated into academic settings to enhance learning experiences.

Table 2 presents the level of technological acceptability among the 1st, 2nd, and 3rd-year undergraduate students, classified into 6 different categories: (a) autonomy, (b) competence, (c) relatedness, (d) perceived ease of use, (e) perceived usefulness, and (f) behavioral intention to use. Based on the results, it can be noted that the standard deviation of all the categories ranged from 0.72, which was less than 1.0, indicating consistency of the responses from the respondents. More so, the level of technological acceptability is high, with an overall mean of 3.11. This indicates high technological acceptability among respondents. This suggests that respondents find ProTox 3.0 to be effective and acceptable and well-accepted among undergraduate students, which would greatly benefit the integration of this technology in academic settings and further enhance learning experiences.

The result is attuned to the study conducted by Castro et al. (2025), which found that the implementation of virtual laboratories in academia made significant positive impacts on the learners' achievement in Chemistry, as per previously conducted meta-analyses about its effects on academic achievement across Science subjects. Additionally, the results support the findings of Chan et al. (2021) that the use of virtual laboratories for Chemistry education can provide better results in cognitive, affective, and skill-based learning than traditional passive media. Since this study focuses more on the affective learning outcomes of ProTox 3.0, this includes their attitude towards chemistry and its usability in terms of satisfaction, usefulness, ease, time efficiency, and realism/authenticity, and self-efficacy through the respondents' confidence

Table 2.*Level of technological acceptability*

A. Autonomy	Mean	SD	Interpretation
1. I feel a sense of choice and freedom using ProTox 3.0.	3.33	0.61	Very High
2. ProTox 3.0 education provides me with interesting options and choices.	3.50	0.59	Very High
3. I have more control while using ProTox 3.0.	3.24	0.66	High
4. ProTox 3.0 gives me more chances to control my tasks.	3.36	0.61	Very High
Category Mean	3.36	0.62	Very High
B. Competence	Mean	SD	Interpretation
1. I am better in ProTox 3.0 than other users.	2.40	0.74	Low
2. I have stronger capability than other users due to ProTox 3.0.	2.63	0.77	High
3. I am superior to others through using ProTox 3.0.	2.21	0.83	Low
4. After receiving a ProTox 3.0 training, I felt competent.	2.95	0.72	High
5. I have learned an interesting new skill through ProTox 3.0.	3.19	0.69	High
Category Mean	2.68	0.83	High
C. Relatedness	Mean	SD	Interpretation
1. I like the ProTox 3.0 users.	3.11	0.54	High
2. ProTox 3.0 gives me more chances to interact with others.	3.08	0.63	High
3. I feel close to others while using ProTox 3.0.	2.84	0.62	High
Category Mean	3.01	0.61	High

D. Perceived Ease of Use	Mean	SD	Interpretation
1. My interaction with ProTox 3.0 solutions is clear and stable.	3.12	0.54	High
2. It is easy for me to become skillful at using ProTox 3.0.	3.09	0.66	High
3. I find ProTox 3.0 easy to use.	3.11	0.64	High
Category Mean	3.10	0.61	High

E. Perceived Usefulness	Mean	SD	Interpretation
1. Using ProTox 3.0 enhances my effectiveness.	3.26	0.56	Very High
2. ProTox 3.0 is useful for my chosen college program.	3.43	0.59	Very High
3. Using ProTox 3.0 increases my productivity.	3.16	0.56	High
Category Mean	3.28	0.58	Very High

F. Behavioral Intention to Use	Mean	SD	Interpretation
1. I intend to use ProTox 3.0 in the future.	3.38	0.61	Very High
2. I plan to use ProTox 3.0 in the future.	3.40	0.60	Very High
3. I predict I will use ProTox 3.0 in the future.	3.41	0.59	Very High
Category Mean	3.40	0.60	Very High
Overall Mean	3.11	0.72	High

in laboratory work, and performance assessment in terms of knowledge gain and log data assessment. Notably, the results were highly positive, reflecting the same results of the effectiveness of virtual laboratories like ProTox 3.0, as an instructive and supplementary tool in teaching.

Autonomy. The items pointed out that the respondents had a very high level of autonomy, with a mean of 3.36, meaning that students perceived a high degree of freedom and control when using the software.

The table further reveals that the mean rating of the items ranges from 3.36 to 3.50. Specifically, the item on “I feel a sense of choice and freedom when using ProTox 3.0” has a mean rating of 3.33, described as Very High, which means that the respondents have the option to utilize the software with creative freedom, while “ProTox 3.0 gives me more chances to control my tasks” with a mean of 3.36, described as Very High. This means that strategic lesson planning that enhances teachers' instruction is often observed. This further means that the findings of this investigation highlight the importance of autonomy and creative freedom in utilizing virtual laboratories.

These findings highlight the importance of autonomy and creative freedom in utilizing virtual laboratories, supporting the study by Bortnik et al. (2017), which found that virtual laboratories serve as effective pre-physical laboratory tools by providing access for students to develop skills before conducting hands-on experiments. Noting the results from Table 2 in this study, with the interpretation of “Very High”, this bolsters the previous studies' findings on the effectiveness of E-labs in cases of certain barriers in conducting a physical chemical experiment, distance learning, and its role as a pre-class training tool.

Competence. The items pointed out that students are more likely to be either competent or have an interest in learning while using the website, with a mean of 2.68, which means that the physical barriers are often evident. The table further reveals that the mean rating of the items ranges from 2.21 to 3.19. Specifically, the item “I am superior to others through using ProTox 3.0” has a mean rating of 2.21, described as “Low”, which means that students using ProTox 3.0 for competitive purposes is sometimes evident, while “I have learned an interesting new skill through ProTox 3.0” with a mean of 3.19, described as “High”. This means that using ProTox 3.0 for general educational purposes is highly observed compared to those using ProTox 3.0 for competitive purposes in education.

This finding supports the study of Bazie, H. et al. (2024), which states that users of ProTox 3.0 are predominantly engaged with the

software for general educational purposes, highlighting its value as an effective digital learning tool. The present results align with Bazie et al.'s conclusions, as both sets of data reveal a positive correlation with a "High" indication, suggesting that users demonstrate strong competence in navigating and utilizing the software to enhance their learning experiences. This correlation further implies that the proficiency in using ProTox 3.0 significantly contributes to improved educational outcomes, as it enables users to access, analyze, and apply information more efficiently. Moreover, this alignment reinforces the notion that technological familiarity and skillful engagement with digital platforms like ProTox 3.0 play a crucial role in promoting academic growth, independent learning, and overall digital literacy among users.

Relatedness. The items pointed out that students had high relatedness with an overall mean of 3.01, which means that the students' relatedness using ProTox 3.0 is often evident. The table further reveals that the mean rating of the items ranges from 2.84 to 3.11. Specifically, the item "I like the ProTox 3.0 users" has a mean rating of 3.11, described as high, which means that teachers' technique in gymnastic instruction is sometimes evident, while "I feel close to others while using ProTox 3.0" with a mean of 2.84, also described as high. This means that using ProTox 3.0 can affect the students' relatedness, which improves their social connections.

This finding supports the study of Bazie, H. et al. (2024), which states that the users of ProTox 3.0 consistently emphasize the significance of using the software for social interactions or connections. Noting that the results of both data are positively correlated with the indication of "Very High," proving the relatedness using ProTox 3.0 is correlated with. This further validates the idea that the use of ProTox 3.0 not only promotes communication and engagement within digital communities but also strengthens interpersonal relationships and collaboration among users, thereby proving that the platform's effectiveness is closely tied to its ability to facilitate social connectivity and interaction.

Perceived Ease of Use. The items pointed out that respondents had high perceived ease of use with a mean of 3.10, which signifies that the users find the ProTox 3.0 easy to use and operate. The table further reveals that the mean rating of the items ranges from 3.09 to 3.12. Specifically, the item on “My interaction with ProTox 3.0 solutions is clear and stable” has a mean rating of 3.12, described as high, which means that the respondents find the software is understandable, on the other hand the mean of 3.09 in “It is easy for me to become skillful at using ProTox” points that this software is simple to master. This means that ProTox 3.0 provides a user-friendly view and uncomplicated interaction, promoting ease and confidence among users. The findings show that the system’s accessibility and clarity can ultimately further enhance user mastery. This also means that the simplicity of ProTox 3.0 contributes positively to user satisfaction and manageable system use.

This finding supports the investigation of Bazie et al. (2024), which confirms the role of virtual labs as effective supplements or alternatives to traditional labs, underscoring that the ease of use is helpful in reducing the resources and time needed. In the study that used a quasi-experimental design where they were divided into 3 groups composed of 20 BS chemistry students from Dilla University, the group that used virtual laboratories scored higher in the post-test than the groups that used the lecture-only method. These findings suggest that computer-based laboratories are a viable and effective alternative, and that the perceived ease of use has prevailed over traditional lecture-based methods.

Perceived Usefulness. The items pointed out that respondents had very high perceived usefulness, with a mean of 3.28, which means that they view the ProTox 3.0 as useful and helpful in their specific field. The table further reveals that the mean rating of the items ranges from 3.16 to 3.43. Specifically, the item on “Using ProTox 3.0 increases my productivity” has a mean rating of 3.16, described as high, which means that respondents saw the potential in efficiency of the software in contributing to their progress. Additionally, “ProTox 3.0 is useful for my chosen college program” with a mean of 3.43, is described as very high. This means that the perceived usefulness that points out the respondents’

viewpoint with regard to applicability is very high. This further means that the findings of this investigation highlight the importance of knowing the students' perceptions are pivotal in seeing the potential of ProTox 3.0. While there may be areas for improvement in this software to garner a higher perceived usefulness score, this investigation underscores the significance of knowing the possible benefits by exploring the users' perspective.

This finding supports the study of Morris & Dillon (1997), which demonstrated that technology acceptance models focusing on system usefulness, beyond traditional usability measures, significantly influence user acceptance of software systems. The abstract of the study states that the system usability has a strong bearing and impact in user acceptance.

Behavioral Intention to Use. The items pointed out that respondents had very high behavioral intention to use with a mean of 3.40, which means that the respondents are willing to continue the use of ProTox 3.0. The table further reveals that the mean rating of the items ranges from 3.38 to 3.41. Specifically, the item "I predict I would use ProTox 3.0 in the future" obtained the highest mean of 3.41, while "I intend to use ProTox 3.0 in the future" had the lowest mean of 3.38, both described as very high. This suggests that users show a strong commitment toward the continuous use of the system. This means that users' experience and perceived benefits from ProTox 3.0 create their strong intention to keep using it. The findings highlight the significant role of potential benefit in the future, perceived usefulness, and ease of use in fostering consistent behavior. This further explains that maintaining system usability, ease, and relevance will sustain users' future engagement.

This finding supports the study (Fabrigas & Paglinawan, 2025; Batamuliza et al., 2024), which highlights that such simulations correlate strongly with better conceptual understanding and retention, highlighting their value in addressing complex topics. The said study in Liberty Bible Baptist Academy Inc., in the province of Bukidnon, Philippines, shows a positive correlation between digital simulation exposure and conceptual

understanding in chemistry with the junior high school students as the participants. All of which suggests the significance of using digital technology together with usability to enhance user comprehension, which can also relate to the intention of use.

Table 3.

Significant differences in technological acceptability when grouped according to the respondents' demographic profile

Age Bracket	N	Mean	SD	F-value	p-value	Interpretation
17-18	96	3.08	0.40	2.077	0.107	Not significant
19-20	21	3.25	0.33			
21-22	1	2.43	-			
23 or above	3	3.00	0.58			
Sex	N	Mean	SD	F-value	p-value	Interpretation
Male	24	3.28	0.47	2.428	0.020	Significant
Female	97	3.06	0.37			
Year Level	N	Mean	SD	F-value	p-value	Interpretation
1st Year	108	3.11	0.41	0.390	0.678	Not significant
2nd Year	10	3.05	0.32			
3rd Year	3	3.29	0.51			
College Program	N	Mean	SD	F-value	p-value	Interpretation
BS Biology	1	3.29	-	0.637	0.700	Not significant
BS Chemistry	5	2.89	0.38			
BS Medical Laboratory Science	73	3.13	0.40			
BS Nursing	9	3.03	0.47			
BS Nutrition and Dietetics	4	3.13	0.17			
BS Pharmacy	26	3.06	0.42			
BS Psychology	3	3.35	0.28			

Significant differences in the technological acceptability among the respondents

Table 3 illustrates the significance of the differences in the level of technological acceptability when grouped according to the respondents' demographic profile. As noted by the p-values of each profile, all, except for sex, are greater than 0.05. As such, they are considered not significant. Therefore, there are no significant differences in the levels of sense of validation regardless of the respondents' age bracket, year level, or chosen program. This may imply that the software is accessible to a wide range of users varying in maturity on their chosen program.

Among the demographic profiles, however, there is another implication in the level of technological acceptability when grouped according to the respondents' sex. The p-value for this profile is 0.020, which is less than 0.05. This shows a significant difference in this profile. Thus, the technological acceptability in using ProTox 3.0 differs in males and females. Some implications that involve this data may be due to the increase in female than male respondents, exhibiting that more female students are interested in exploring how ProTox 3.0 works in the laboratory.

Understandably, these findings may align with a study by Sormin et al. (2023), which attempted to gain insights into male and female students in utilizing virtual chemistry laboratories, a software similar to how ProTox 3.0 works. In that study, they found that males' views on virtual laboratories are significantly larger than those of females, noting that the mechanics involved work similarly to playing online games, which may contribute to why males favor it more than females. In another study by Oladejo et al. (2021) conducted at Lagos State University, they identified that to improve these insights, allowing both genders to expose themselves to computer simulations increases competency for chemistry courses. Further, the results show no significant difference in gender when this treatment is given, meaning that exposure can be an applied method for all to improve.

CONCLUSION AND RECOMMENDATIONS

Conclusions

After analyzing the results of the study's respondents, variables, and their corresponding indicators, as well as the comparisons and correlations present in the data, the researchers have drawn the following conclusions regarding the qualitative behaviors of each problem statement.

1. The profile of the respondents is mostly aged 17-18, with a noticeable female majority and more 1st Year students than 2nd and 3rd Year students. The largest group by college program belongs to BS Medical Laboratory Science, followed by BS Pharmacy, BS Nursing, BS Chemistry, BS Nutrition and Dietetics, BS Psychology, and then BS Biology. This demographic profile sets the stage for further analysis of the versatility of ProTox 3.0 and its technological acceptance.
2. The level of technological acceptability among the respondents when using ProTox 3.0 as a predictive chemical toxicity software is high. In particular, the level of the respondents' autonomy and perceived usefulness of the software is very high. Meanwhile, their level of competence, relatedness, perceived ease of use, and behavioral intention to use the software is high. As such, increasing their exposure to ProTox 3.0 can help improve their intrapersonal and interpersonal experiences in using related chemistry software, creating an environment where chemical data can be easily derived via reliable *in silico* models.
3. There are no significant differences in the level of technological acceptability when grouped according to respondents' age bracket, year level, and college program. However, there is a significant difference when it is grouped by their sex. Hence, this can imply that there is a need to balance and equalize the experiences for both sexes to holistically improve the quality of using ProTox 3.0. In

contrast, enriching the impact of this software for laboratory experimentation requires more specific factors to yield similar improvements in their work, rather than finding patterns in their demographic profiles.

Recommendations

In relation to the formulated conclusions, the researchers propose the following recommendations when conducting and developing future research targeted to similar topics.

1. Future research should explore wider demographic and diverse participants, not only with students and professionals, but also people in marginalized and underrepresented communities, to reflect diverse realities, especially in areas of user acceptance.
2. Cross-verify the software toxicity prediction with actual laboratory or experiment results, fostering a culture of validity and increasing the knowledge of individuals using tangible laboratory apparatus and chemicals, relating to learners who prefer visual and actual learning, strengthening the integration of traditional laboratory works and technological advancement.
3. Include interactive visualizations, tutorials, and infographics to further simplify and promote the use of the software for future respondents in similar research.
4. Perform similar research on a narrower scope of respondents, such as focusing on a specific college program or year level, to gain a more comprehensive understanding of the population's insights on using ProTox 3.0.
5. Conduct research that highlights other key aspects of ProTox 3.0, such as the utility of toxicology reports and the impact of using the software in an actual laboratory experiment.

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APPENDICES

APPENDIX A

LETTERS OF APPROVAL

August 28, 2025

DR. KATHLEEN B. OLAÑO, RPh
Dean, College of Pharmacy and Chemistry
University of the Immaculate Conception
Fr. Selga St., Davao City, Philippines

Dear Dr. Olaño:

Praised be Jesus and Mary!

We, the first-year students of the Bachelor of Science in Chemistry program, are currently writing our research as one of the requirements for GEC003.23: Science, Technology, and Society. With that, we are working on a study titled **"STUDENTS' LEVEL OF TECHNOLOGICAL ACCEPTABILITY ON THE USE OF PREDICTIVE CHEMICAL TOXICITY SOFTWARE FOR SCIENCE-RELATED UNDERGRADUATE PROGRAMS IN A CATHOLIC UNIVERSITY IN DAVAO CITY."**

This study aims to measure the level of technological acceptability when using a chemical toxicity prediction model, such as ProTox 3.0, of our chosen respondents - the students in the University of the Immaculate Conception studying BS Pharmacy, BS Chemistry, BS Nutrition and Dietetics, BS Medical Laboratory Science, BS Biology, and BS Psychology - to better understand the usability of this software and understand its potential in chemical research, safety, and experimentation. As researchers, we seek to address the issue of toxicity hazards in undergraduate laboratories, especially in identifying what measures may be taken to help by thoroughly testing their insights on the study's problems at hand.

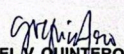
In line with this, we would like to ask for your office's permission for us to conduct the study. As the nature of our research is quantitative, the data collection will gather our respondents' insights on the research topic to understand the levels and relationship with the respondents and the variable presented in this study, benefiting the researchers, students, faculty, and the academic community, as it will enhance the knowledge surrounding chemical safety and how integrating predictive software may play a role in it. Specifically, we will administer the data collection process by providing the respondents a brief orientation on using the ProTox software, then collect quantitative data using a survey questionnaire, which will be distributed to the respondents on September 8-12, 2025. Rest assured that safety will be our utmost priority within this study, adhering to safety regulations and other measures, and obtaining the necessary consent and guidance. With this, we are further aware that the school may not be held liable or accountable for any unanticipated incidents that may happen, and we will treat this obligation seriously.

We are hoping for your positive response regarding this request.

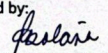
Respectfully yours,


GABRIEL ANTONI B. LAGURA
Research Representative

Noted by:


MS. GENEVELYN QUINTERO, LPT
GEC003.23: Science, Technology, and Society Instructor

Approved by:

 8/28/2025
DR. KATHLEEN B. OLAÑO, RPh
Dean, College of Pharmacy and Chemistry

September 2, 2025

MR. CEASAR IAN P. BENABLO
Dean, College of Computer Studies
University of the Immaculate Conception
Fr. Seiga St., Davao City, Philippines

Dear Mr. Benablo:

Praised be Jesus and Mary!

We, the first-year students of the Bachelor of Science in Computer Science program, are currently writing our research as one of the requirements for GEC003.23: Science, Technology, and Society. With that, we are working on a study titled **"STUDENTS' LEVEL OF TECHNOLOGICAL ACCEPTABILITY ON THE USE OF PREDICTIVE CHEMICAL TOXICITY SOFTWARE FOR SCIENCE-RELATED UNDERGRADUATE PROGRAMS IN A CATHOLIC UNIVERSITY IN DAVAO CITY."**

This study aims to measure the level of technological acceptability when using a chemical toxicity prediction model, such as ProTox 3.0, of our chosen respondents - the students in the University of the Immaculate Conception studying BS Pharmacy, BS Chemistry, BS Nutrition and Dietetics, BS Medical Laboratory Science, BS Biology, and BS Psychology - to better understand the usability of this software and understand its potential in chemical research, safety, and experimentation. As researchers, we seek to address the issue of toxicity hazards in undergraduate laboratories, especially in identifying what measures may be taken to help by thoroughly testing their insights on the study's problems at hand.

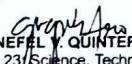
In line with this, we would like to ask for your office's permission for us to conduct the study. As the nature of our research is quantitative, the data collection will gather our respondents' insights on the research topic to understand the levels and relationship with the respondents and the variable presented in this study, benefiting the researchers, students, faculty, and the academic community, as it will enhance the knowledge surrounding chemical safety and how integrating predictive software may play a role in it. Specifically, we will administer the data collection process by providing the respondents a brief orientation on using the ProTox software; then collect quantitative data using a survey questionnaire, which will be distributed to the respondents on September 8-12, 2025. Rest assured that safety will be our utmost priority within this study, adhering to safety regulations and other measures, and obtaining the necessary consent and guidance. With this, we are further aware that the school may not be held liable or accountable for any unanticipated incidents that may happen, and we will treat this obligation seriously.

We are hoping for your positive response regarding this request.

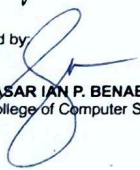
Respectfully yours,


ALTHEA DENULLE M. JAVILLO
Research Representative

Noted by:


MS. GENEVEL Y. QUINTERO, LPT
GEC003.23 Science, Technology, and Society Instructor

Approved by:


MR. CEASAR IAN P. BENABLO
Dean, College of Computer Studies

September 18, 2025

DR. NELCA STEPHANY U. AMATONG, RND
Dean, College of Human Environmental and Food Sciences
University of the Immaculate Conception
Fr. Selga St., Davao City, Philippines

Dear Dr. Amatong:

Praised be Jesus and Mary!

We, the first-year students of the Bachelor of Science in Chemistry program, are currently writing our research as one of the requirements for GEC003.23: Science, Technology, and Society. With that, we are working on a study titled **"FIRST- TO THIRD-YEAR STUDENTS' LEVEL OF TECHNOLOGICAL ACCEPTABILITY ON A PREDICTIVE CHEMICAL TOXICITY SOFTWARE FOR SCIENCE-RELATED UNDERGRADUATE PROGRAMS IN A CATHOLIC SCHOOL IN DAVAO CITY."**

This study aims to measure the level of technological acceptability when using a chemical toxicity prediction model, such as ProTox 3.0, of our chosen respondents - the students in the University of the Immaculate Conception studying BS Pharmacy, BS Chemistry, BS Nutrition and Dietetics, BS Medical Laboratory Science, BS Biology, and BS Psychology - to better understand the usability of this software and understand its potential in chemical research, safety, and experimentation. As researchers, we seek to address the issue of toxicity hazards in undergraduate laboratories, especially in identifying what measures may be taken to help by thoroughly testing their insights on the study's problems at hand.

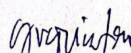
In line with this, we would like to ask for your office's permission for us to conduct the study in your college. As the nature of our research is quantitative, the data collection will gather our respondents' insights on the research topic to understand the levels and relationship with the respondents and the variable presented in this study, benefiting the researchers, students, faculty, and the academic community, as it will enhance the knowledge surrounding chemical safety and how integrating predictive software may play a role in it. Specifically, we will administer the data collection process by providing an online survey questionnaire that briefs the respondents on using the ProTox software, then collect quantitative data thereafter, which will be distributed to the respondents on September 18-22, 2025. Rest assured that safety will be our utmost priority within this study, with necessary consent and guidance, and keeping the respondents' record confidential as part of the Data Privacy Act of 2012. With this, we are further aware that the school may not be held liable or accountable for any unanticipated incidents that may happen, and we will treat this obligation seriously.

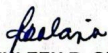
We are hoping for your positive response regarding this request.

Respectfully yours,

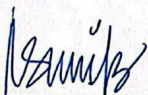

GABRIEL ANTON B. LAGURA
Research Representative

Noted by:


MS. GENEVEL V. QUINTERO, LPT
Instructor, GEC003.23: Science, Technology, and Society


DR. KATHLEEN B. OLAÑO, RPh
Dean, College of Pharmacy and Chemistry

Approved by:


DR. NELCA STEPHANY U. AMATONG, RND
Dean, College of Human Environmental and Food Sciences

APPENDIX B

SURVEY QUESTIONNAIRE

Dear Respondent,

The researchers will conduct a study on your Level of Technological Acceptability upon using ProTox 3.0. As such, your participation in answering this survey, as part of their study, is highly appreciated. The researchers request that you answer the items honestly and holistically, as the study's success will depend on your responses, which will be confidential and only be used for research purposes. Thank you for your time and valuable contribution.

Respectfully yours,
The Researchers

Part I. Profile of the Respondents

Instructions: Properly shade the circle that fits your information below.

Age: ☐ 17-18 ☐ 19-20 ☐ 21-22 ☐ 23 above	
Sex: ☐ Male ☐ Female	Year Level: ☐ 1st Year ☐ 3rd Year ☐ 2nd Year
Program: ☐ Bachelor of Science in Biology ☐ Bachelor of Science in Chemistry ☐ Bachelor of Science in Medical Laboratory Science ☐ Bachelor of Science in Nursing ☐ Bachelor of Science in Nutrition and Dietetics ☐ Bachelor of Science in Pharmacy ☐ Bachelor of Science in Psychology	

Part II. What is ProTox 3.0?

Before answering the survey, the researchers would like you to watch this video to understand what ProTox 3.0 does as a predictive chemical toxicity software.

ProTox 3.0 is a computer tool used to predict how harmful small molecules might be based on their structures. It is helpful in environmental studies, checking chemical safety, and early drug discovery. By looking at a molecule's structure, **ProTox 3.0** can predict if it may cause problems like acute toxicity (bad effects right away), mutagenicity (could damage genes), carcinogenicity (might cause cancer), or harm to the environment.

It works by comparing the molecule's structure to a big database of known harmful features. Scientists and regulators find it useful because they can input a molecule's details and get a clear report about its potential risks. This helps them make decisions without needing as many expensive and time-consuming lab tests.

Source:

ProTox 3 Tool (2024 June 25). ThePharmaLab.
https://www.youtube.com/watch?v=DZQTb54__hk

Watch the video before proceeding with the survey, and be sure to experiment with inputting other compounds once you have finished the video.

Part III. Level of Technological Acceptability

Instructions: Please rate the items below according to your level of agreement for each statement by placing a check mark (✓) on the corresponding box using the following Likert scale:

4 - Strongly Agree | 3 - Agree | 2 - Disagree | 1 - Strongly Disagree

ITEM	RATING			
	4	3	2	1
A. Autonomy				
1. I feel a sense of choice and freedom using ProTox 3.0.				
2. ProTox 3.0 education provides me interesting options and choices.				
3. I have more control while using ProTox 3.0.				
4. ProTox 3.0 gives me more chances to control my tasks.				
B. Competence	4	3	2	1
1. I am better in ProTox 3.0 than other users.				
2. I have stronger capability than other users due to ProTox 3.0.				
3. I am superior to others through using ProTox 3.0.				
4. After receiving a ProTox 3.0 training, I felt competent.				
5. I have learned an interesting new skill through ProTox 3.0.				
C. Relatedness	4	3	2	1
1. I like the ProTox 3.0 users.				
2. ProTox 3.0 gives me more chances to interact with others.				
3. I feel close to others while using ProTox 3.0.				
4. I have more opportunities to be close to others through ProTox 3.0.				
D. Perceived Ease of use	4	3	2	1
1. My interaction with ProTox 3.0 solutions is clear and stable.				
2. It is easy for me to become skillful at using ProTox 3.0.				

3. I find ProTox 3.0 easy to use.				
E. Perceived Usefulness	4	3	2	1
1. Using ProTox 3.0 enhances my effectiveness.				
2. ProTox 3.0 is useful for my chosen college program.				
3. Using ProTox 3.0 increases my productivity.				
F. Behavioral Intention to Use	4	3	2	1
1. I intend to use ProTox 3.0 in the future.				
2. I plan to use ProTox 3.0 in the future.				
3. I predict I would use ProTox 3.0 in the future.				

Source: Racero, F.J., Bueno, S., & Gallego, M.D. (2020). *Predicting Students' Behavioral Intention to Use Open Source Software: A Combined View of the Technology Acceptance Model and Self-Determination Theory*. MPDI Applied Sciences. <https://doi.org/10.3390/app10082711>

APPENDIX C

INFORMED CONSENT

Informed Consent Form for: First- to Third-Year Students' Level
of Technological Acceptability on the
Use of Predictive Chemical Toxicity
Software for Science-Related
Undergraduate Programs in a Catholic
School in Davao City

Name of the Researcher(s): Students of BSCS-1A and BSChem-1A

Institution: University of the Immaculate Conception

INTRODUCTION

You are invited to participate in a research study conducted by the students of BSCS-1A and BSChem-1A, at the University of the Immaculate Conception, because you fit the inclusion criteria as an informant of this study.

Your participation is completely voluntary. Please read the information below, and ask questions about anything you do not understand, before deciding whether to participate. Please take as much time as you need to read the consent form. You may also decide to discuss participation with your family or friends.

If you decide to participate, you will be asked to answer this form. You will be given a copy of your responses once you submit them.

PURPOSE OF THE STUDY

This study aims to determine the levels of technological acceptability when using predictive chemical toxicity software among first- to third-year students under science-related undergraduate programs. Furthermore, it also measures if there exists a significant difference between the level of technological acceptability among the first- to third-year students.

STUDY PROCEDURES

To ensure transparency, this survey will provide a brief orientation regarding the purpose and requirements of this study, as formulated by

the researchers. After this, your consent will be asked through this consent form. In order to obtain reliable and generalized results for this study, the researchers aim to gather 200 respondents who fit their inclusion criteria. Alongside this, if you give your consent to the researchers to participate in this study, you will be asked to answer a survey questionnaire about this research topic. This procedure will only take 15-20 minutes to complete.

In addition, to be able to answer the survey, the researchers would like to require you to have a stable internet connection as well as access to your device's web browser.

POTENTIAL RISKS AND DISCOMFORTS

The provided survey consists of questions that are not entirely sensitive but may cause discomfort. To address this, the researchers will guide you and answer any queries or concerns you may have about the survey items, or the research itself. You may also opt not to answer the survey if you feel psychological or emotional distress, or you can withdraw as a respondent in the study if you acknowledge that you cannot share the information that is asked of you. The researchers value your participation and will place your welfare as their highest priority throughout the study.

POTENTIAL BENEFITS TO PARTICIPANTS AND/OR TO SOCIETY

Through proper statistical and systemic analysis, the possible results of this study could provide benefits to the following beneficiaries. This study will help students understand the importance and applicability of predictive chemical toxicity software to enhance their knowledge of computational toxicology essential for modern scientific research. Additionally, this study provides educators with insights into students' acceptance and adaptability towards integrating predictive chemical toxicity software into the curriculum. Educators can also use these findings to improve teaching methods, implement technology-based activities, and prepare students for advanced scientific roles.

This study also provides Catholic institutions and their stakeholders with valuable insights into adopting advanced technological tools for enhancing science education and research capabilities in line with modern trends. Lastly, insights from the study could help software developers of predictive chemical toxicity software and its related technology understand the user acceptance challenges and requirements from an academic

perspective, facilitating improvements and wider adoption.

CONFIDENTIALITY

We will keep your records for this study confidential in coordination with the Data Privacy Act of 2012. Any identifiable information obtained in connection with this study will remain confidential, except if necessary to protect your rights or welfare. This certification means that the researcher can resist the release of information about your participation to people who are not connected with the study. When the results of the research are published or discussed at conferences, no identifiable information will be used, regardless of institution, organization, or the names of the respondents involved.

PARTICIPATION AND WITHDRAWAL

Your participation is voluntary. Your refusal to participate will involve no penalty or loss of benefits to which you are otherwise entitled. You may withdraw your consent at any time and discontinue participation without penalty. You are not waiving any legal claims, rights, or remedies because of your participation in this research study. Rest assured that your withdrawal will prevent us from using your response in our study.

INVESTIGATOR'S CONTACT INFORMATION

If you have any questions or concerns about the research, please feel free to contact the researcher at the University of the Immaculate Conception - Main Campus, Fr. Selga St., Davao City, or through email via glagura_250000000840@uic.edu.ph.

RIGHTS OF RESEARCH RESPONDENTS

If you have questions, concerns, or complaints about your right as a research participant, or the research in general, and are unable to contact the research team, or if you want to talk to someone independent of the research team, please contact the University of the Immaculate Conception at (082) 221 8090.

RESEARCH RESPONDENT'S CONSENT

I have read the information provided above. I have been given a chance to ask questions. My questions have been answered to my satisfaction, and I agree to participate in this study. I have been given a copy of this form. I can withdraw my consent at any time and discontinue participation without penalty.

APPENDIX D

SPSS RESULTS

AgeBracket

	Frequency	Percent	Valid Percent	Cumulative Percent
Valid 17-18 years old	96	79.3	79.3	79.3
19-20 years old	21	17.4	17.4	96.7
21-22 years old	1	.8	.8	97.5
23 years old or above	3	2.5	2.5	100.0
Total	121	100.0	100.0	

Sex

	Frequency	Percent	Valid Percent	Cumulative Percent
Valid Male	24	19.8	19.8	19.8
Female	97	80.2	80.2	100.0
Total	121	100.0	100.0	

YearLevel

	Frequency	Percent	Valid Percent	Cumulative Percent
Valid 1st Year	108	89.3	89.3	89.3
2nd Year	10	8.3	8.3	97.5
3rd Year	3	2.5	2.5	100.0
Total	121	100.0	100.0	

CollegeProgram

	Frequency	Percent	Valid Percent	Cumulative Percent
Valid Bachelor of Science in Biology	1	.8	.8	.8
Bachelor of Science in Chemistry	5	4.1	4.1	5.0
Bachelor of Science in Medical Laboratory Science	73	60.3	60.3	65.3
Bachelor of Science in Nursing	9	7.4	7.4	72.7
Bachelor of Science in Nutrition and Dietetics	4	3.3	3.3	76.0
Bachelor of Science in Pharmacy	26	21.5	21.5	97.5
Bachelor of Science in Psychology	3	2.5	2.5	100.0
Total	121	100.0	100.0	

Descriptives

TechnologicalAcceptability

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
17-18 years old	96	3.0848	.40159	.04099	3.0034	3.1662	1.90	4.00
19-20 years old	21	3.2500	.33174	.07239	3.0990	3.4010	2.48	3.86
21-22 years old	1	2.4300	.	.	.	.	2.43	2.43
23 years old or above	3	3.0000	.58198	.33601	1.5543	4.4457	2.62	3.67
Total	121	3.1060	.39968	.03633	3.0340	3.1779	1.90	4.00

ANOVA

TechnologicalAcceptability

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.969	3	.323	2.077	.107
Within Groups	18.200	117	.156		
Total	19.169	120			

Group Statistics

	Sex	N	Mean	Std. Deviation	Std. Error Mean
TechnologicalAcceptability	Male	24	3.2750	.47396	.09675
	Female	97	3.0641	.37006	.03757

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
TechnologicalAcceptability	Equal variances assumed	2.428	.122	2.358	119	.020	.21088	.08944	.03378	.38797
	Equal variances not assumed			2.032	30.297	.051	.21088	.10379	-.00100	.42275

Descriptives

TechnologicalAcceptability

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
1st Year	108	3.1058	.40555	.03902	3.0285	3.1832	1.90	4.00
2nd Year	10	3.0530	.32108	.10153	2.8233	3.2827	2.48	3.67
3rd Year	3	3.2867	.50836	.29350	2.0238	4.5495	2.71	3.67
Total	121	3.1060	.39968	.03633	3.0340	3.1779	1.90	4.00

ANOVA

TechnologicalAcceptability

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.126	2	.063	.390	.678
Within Groups	19.043	118	.161		
Total	19.169	120			

Descriptives

TechnologicalAcceptability

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Bachelor of Science in Biology	1	3.2900	.	.	.	.	3.29	3.29
Bachelor of Science in Chemistry	5	2.8860	.37997	.16993	2.4142	3.3578	2.48	3.48
Bachelor of Science in Medical Laboratory Science	73	3.1333	.39854	.04665	3.0403	3.2263	2.19	4.00
Bachelor of Science in Nursing	9	3.0311	.47115	.15705	2.6690	3.3933	2.38	3.76
Bachelor of Science in Nutrition and Dietetics	4	3.1325	.16998	.08499	2.8620	3.4030	3.00	3.38
Bachelor of Science in Pharmacy	26	3.0581	.42487	.08332	2.8865	3.2297	1.90	3.81
Bachelor of Science in Psychology	3	3.3500	.28160	.16258	2.6505	4.0495	3.14	3.67
Total	121	3.1060	.39968	.03633	3.0340	3.1779	1.90	4.00

ANOVA

TechnologicalAcceptability

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.622	6	.104	.637	.700
Within Groups	18.547	114	.163		
Total	19.169	120			